Phase and activity checks

Work individually. Follow the elapsed times in the slides. Keep original answers and add corrections. Use extra paper where needed.

A. Phase and material balance (15–35 min)

Ideal mixture at 350 K: z_1=0.5, P_1^{sat}=150 kPa, P_2^{sat}=60 kPa.

A1. Calculate P_{bubble} and P_{dew}. Show substitution.

A2. Identify each phase state and compare with the boundary pressures.

P / kPa	Phase state	Reason
80		
95		
110		

A3. At 110 kPa: β=___ x_1=___ Does a vapor composition exist? Explain.

A4. At 95 kPa use x_1=0.388889, y_1=0.614035, β=0.493506. Substitute into $z_1=(1-\beta)x_1+\beta y_1$:

x_2=___ y_2=___ x_1+x_2=___ y_1+y_2=___

B. Inferred gamma and activity (48–65 min)

P=100 kPa, x_1=0.4, y_1=0.6. Assume ideal vapor and negligible Poynting correction.

B1. With P_1^{sat}=120 kPa, show $\gamma_1=y_1P/(x_1P_1^{sat})$ and $a_1=\gamma_1x_1$.

B2. Repeat using P_1^{sat}=150 kPa.

B3. Which input changed? Did the observations change? Can fitting alone establish which vapor pressure is correct?

Vapor-pressure conventions

C. Standalone Antoine exercise (72–90 min)

Use only this synthetic correlation, valid at 20–80 °C:
$\log_{10}(P^{sat}/kPa)=4-1000/(T_C+200)$. Do not use it as Lab 08 input.

C1. Calculate P^{sat} at 50 °C with substitution.

C2. Why is entering 323.15 directly for T_C incorrect?

C3. Can you use the same constants unchanged if $\log_{10}$ becomes ln?

C4. Can you use the correlation at 100 °C? Explain.

D. Lab 08 property check (100–120 min)

Open Lab 08. Load its synthetic example and validate it. Use its own constants for this section.

D1. Baseline file name: _____
Measured-data units: T=_____ P=_____

D2. For component 1, record the exact Antoine equation and log base.

A=_____ B=_____ C=_____
Equation T unit=_____ P^{sat} unit=_____
Valid range: _____ to _____ K (Lab range fields use kelvin).

D3. Select one interior row. Row identifier=_____ T=_____ K.
Convert T to the equation unit. Show the P_1^{sat} calculation and convert to kPa.

P_1^{sat} by hand=_____ kPa. Continue with the same row on page 3.

Lab evidence and interpretation

E1. Independent gamma check (110–120 min)

Use the same component and row as D3, page 2.

T=_____ K P=_____ kPa x_1=_____ y_1=_____

Show $\gamma_1 = y_1 P/(x_1 P_1^{sat})$, including substitution and units:

γ_1 by hand=_____ Table ln γ_1=_____
exp(table ln γ_1)=_____ Difference/reason (allow for rounding):

E2. Two model fits (120–135 min)

Keep observations, properties and fitting objective unchanged. Use NRTL α=0.3 with temperature dependence off. After each fit select **View: Pressure and vapor residuals**.

Model	Mean absolute pressure error (%)	Mean absolute y error (percentage points)
Margules, 1 parameter		
NRTL, fixed α=0.3		

Residual pattern for Margules:

Residual pattern for NRTL:

Any parameter-bound warnings:

Save study JSON and residuals CSV for each model before switching.
Margules file names: _____
NRTL file names: _____

F. Interpretation (135–145 min)

State a finding, give numerical evidence from E2 and explain one limitation. Distinguish reproduction of fitted data from consistency and prediction.

Keep this page available for review question Q6 and the exit ticket.

Review test and corrections

145–172 min. Read the questions in the slides titled *Review test: phase and activity* (Q1–Q3) and *Review test: properties and evidence* (Q4–Q6). Write answers here before viewing the solutions.

Use your calculator and lesson notes. Attendance credit is for completing the activity, not a minimum accuracy score.

Q1. Phase state and boundary calculation

Q2. Vapor fraction from material balance

Q3. Gamma and activity, with substitution

Q4. Antoine calculation and temperature units

Q5. What does lower training error establish?

Q6. Your Lab comparison and its limitation

By study minute 172, **submit Q1–Q6 for attendance**: enter your student ID/name and transfer answers with calculations and corrections. Save the receipt. Retain pages 1–4 and Lab exports.

At 172–180 min, submit the anonymous Session 2 exit ticket and feedback. This form does not identify attendance.